

1. Select 3 different financial assets with at least 3 years of historical data.
2. Implement time series decomposition:
  - Break down each asset's price series into trend, seasonal, and residual components
  - Create visualizations showing each component separately
  - Analyze whether seasonality exists and on what timeframe (daily, weekly, monthly)
  - Calculate the relative contribution of each component to overall price movement
3. Develop and compare multiple forecasting approaches:
  - Implement a simple moving average forecast model
  - Create an ARIMA model with optimized parameters
  - Apply exponential smoothing methods
  - Implement at least one additional method of your choice
4. Build an evaluation framework:
  - Split data into training and testing periods
  - Generate forecasts for different time horizons (1-day, 1-week, 1-month)
  - Calculate accuracy metrics (RMSE, MAE, directional accuracy)
  - Create visualizations comparing predicted vs. actual values
  - Analyze which assets and time periods are most predictable
5. Design a signal generation component: